

Challenges on the Brazilian Information Systems Education: The Professors' Perspective

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Abstract—Contribution: The main contribution of this study is to gather and report on the challenges faced by IS courses based on the professors' perspectives.

Background: Information systems (ISs) area focuses on the interplay among people, processes/organizations, and technologies. As such, stimulating technical and nontechnical competencies are required from professors and students. For instance, despite the current number of IS bachelor's degree courses, several difficulties have been faced, e.g., students dropout, novel careers in Computing and Informatics, and market waves.

Research Questions: What are the challenges and difficulties of IS Education, considering the professors' perspective?

Methodology: A survey research was executed to collect data related to the IS courses, and a qualitative analysis was performed to answer the research question based on coding procedures. The nature of the research is exploratory and descriptive.

Findings: Results reveal the importance of continuous training for professors, the institutional policies to support students, the creation of hybrid courses to reduce barriers to students, and course class time offered in the mornings and afternoons.

Index Terms—Grounded theory (GT), higher education, information systems (ISs), survey.

I. INTRODUCTION

INFORMATION systems (ISs) are a cornerstone in the routine of modern companies. IS collect, store, process, and deliver strategic information to support companies in the decision-making processes. Then, the training of professionals with specific abilities to engineer and manage those systems is still of paramount importance. In Brazil, for historical reasons,

IS courses are majorly proposed and managed by Computing faculty colleges. Brazil offers 791 ongoing IS courses in private and public institutions.¹ The Brazilian Computer Society (SBC) is the entity that interfaces the communication with the Ministry of Education to establish the guidelines followed by the Education institutions to structure their IS courses.

However, IS bachelor's degrees in Brazil have suffered several problems and faced challenges, such as: 1) students dropout (41.7%) once they seek other related courses, such as software engineering (SE); 2) an increasing lower demand (23.4%) for students to the detriment of recently created courses that attract students due to their innovative appeal, such as Artificial Intelligence and Data Science; and 3) a low understanding of the strategic importance of IS professionals and of the differences between the IS career and other Computing careers, and other reasons that have affected the demand for IS courses and the students' permanence [1], [2].

The five authors of this article are professors in IS courses at Brazilian public universities, located in three out of five national regions. In addition to the issues mentioned above, the authors have also faced problems, such as students who do not participate appropriately in classes, students that are only interested in tools used in the industry and not in learning the necessary fundamental concepts, and lack of a proper background from some students. Therefore, the problem addressed in this research is identifying the factors that affect the maintenance of IS courses in Brazil. The goal is to better understand the challenges faced by professors in different regions of the country and how these obstacles impact the quality of education provided to students. By investigating the extent and prevalence of these issues nationwide, government and university managers can develop policies and strategies to enhance the quality of IS teaching and overcome the actual difficulties faced by students and professors.

Based on a literature review, some studies have investigated the IS teaching scenario from the student's perspective [1], [3], [4], [5], [6]. Studies investigating the professor's perspective focus mainly on issues involving curricular adequacy [7], [8] or concerns involving a specific topic in IS [9], [10], [11], [12], [13], [14]. However, from an investigation in the literature, only the study [15] investigates the professor's perspective to characterize the IS faculty at American universities, but it is over 20 years old. Thus, recent research investigating the challenges and difficulties of IS education in Brazil

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¹Search in <https://emec.mec.gov.br/> in 2 November 2022.

from the professors' perspective is lacking. Therefore, this article presents an exploratory study regarding IS education, and the main research question addressed is: "What are the challenges and difficulties of IS Education in Brazil, considering the professors' perspective?". This research is necessary to understand how education in IS has been conducted in different regions of Brazil and it is a first step in order to improve teaching in IS courses and overcome the real challenges faced by students and professors.

A survey research with IS professors in Brazil was conducted to collect their perceptions on the challenges faced in their contexts and 70 responses from all the Brazilian regions were received. Quantitative and qualitative analyses were performed in order to identify: 1) the main difficulties in teaching IS related to the course, the university, the professor, and the students; 2) similarity among the difficulties faced in other Computing courses; 3) IS-related topics that are difficult to teach; and 4) characteristics of a good professor in IS courses. Results reveal the importance of continuous training for professors, institutional policies to support students, creation of hybrid courses to reduce barriers to students, and course class time offering in the mornings and afternoons.

This work contributes by identifying various challenges IS professors face, including difficulties related to the professor, students, course, and university. Additionally, the authors identify topics that are particularly challenging to teach, as well as the characteristics required for an effective IS professor. Furthermore, this work proposes several strategies to address these challenges, including closer collaboration with industry partners, the adoption of innovative teaching methodologies to motivate students, improved training for professors, increased promotion of the IS course, strengthening of basic education, and the creation of hybrid courses. These policies and actions can help mitigate the issues identified and improve IS education quality.

The remainder of this article is structured as follows: Section II presents related work; Section III explains the methodology used to carry out the study; Section IV presents the results; Section V brings the discussion and implications; and Section VI concludes this article with final remarks and future work.

II. RELATED WORK

The ISs undergraduate degree (ISUD) Education has been a matter of investigation since the 1980s in Brazil [16] and in other countries [15], [17], [18]. Some studies report students' points of view on ISUD difficulties from a regional or national perspective [1], [3], [4], [5], [6]. However, most of the works that investigate the professor's view in this context analyze the adequacy of the curriculum of their courses [7], [8], try to understand teaching methodologies [9], [10], [11], [19], [20], and verify if the academic curriculum is aligned with the industry's needs [8].

Some studies investigate the challenges from the professor's perspective but on specific contents in the context of ISUD, such as SE and Computer Programming [12], [13], [14]. Ferreira et al. [12] conducted a survey with SE professors,

including those who teach for IS, to identify emerging higher education topics and difficulties faced in their teaching-learning process. As a result, the authors found challenges related to pedagogy, material resources, students, and content taught. de Lima [13] conducted interviews with professors of the IS course seeking to understand the educational process of computer programming courses. This study had a local scope since only one university's professors participated. Participants mention that technical and pedagogical preparation and practical experience in the area are necessary characteristics for better teaching of programming. However, participants also report that they had no pedagogical preparation for teaching, despite being flexible in improving and applying new learning methods. Berssanette and de Francisco [14] carried out a qualitative exploratory study with 122 professors who teach programming. They found that professors consider it essential to have a teaching practice based on learning theories and that postgraduate training contributes to the teaching profession.

However, the authors affirm that few studies analyze a large-scale panorama of ISUD Education for a specific country and from the professor's perspective on ISUD Education. To their best efforts, the authors found only the work of Gill and Hu [15]. Gill and Hu conducted a survey with professors to understand how IS education was in the U.S. They asked 90 questions to investigate how the IS courses were incorporated in institutions with one or more IS professors, the characteristics of the professors that teach IS, and how IS content evolves to respond quickly to changes. The survey was answered by 240 IS professors from American universities. The authors found that the educational program's quality depends on the faculty's quality. They also found that faculty quality indicators are technical competence in the area taught by the professor and IS-related nonacademic employment experience. Also, IS topics have changed dramatically over time, but the content adopted in the classroom by professors approached the industry demands at the time. It is noteworthy, however, that this work was carried out more than 20 years ago, and, in addition, the focus was to characterize the IS content and faculty. In this work, it was also investigated the main challenges faced by professors in IS teaching.

In summary, the authors are not aware of other studies with: 1) national coverage in Brazil; 2) with the purpose of collecting the challenges faced in ISUD; and 3) reporting evidence found from the professors' perspective. In the next section, it was presented the methodology followed to perform this study.

III. METHODOLOGY

The goal of this study is to analyze the scenario of Education in ISs in Brazil. In order to characterize such a scenario, the authors consider the challenges from IS professors' point of view. To do so, the authors conducted a survey research to collect and explore data related to the IS courses in which the study participants teach. This research is exploratory in nature. Such analysis and results provide to the authors valuable inputs for possible directions of future research [21].

The research philosophical stance of this work is Constructivist/Interpretivist. The authors made reflexive

interactions, analyzing the collected data from participants and discussing it with each other [22]. The discussion aims to reveal important findings from the phenomenon and consider the participant's point of view, especially, about whom, when, where, how, what consequences, and what conditions. In the end, the authors construct a reality from the participant's context. In addition, according to Strauss and Corbin [23], the research question is broad and open-ended, and the literature may be consulted to enhance theoretical sensitivity.

The survey was designed in adequacy to the process defined by Kitchenham and Pfleeger [24] and was divided into four main activities: 1) definition of the study objectives, i: identification of the objectives, and research questions; 2) survey planning: definition of the study execution plan; (3) survey execution: send of the e-mails with the survey questionnaire to the participants for data collection; and 4) analysis and discussion of the data: data extraction, analysis of relevant information for evaluation of the research questions, and discussion of findings.

A. Planning

Initially, the authors designed a questionnaire for professors who have taught in IS courses. This questionnaire was available on Google Forms. It has *characterization questions* to trace the profile of these professors as well as the courses they teach to ISUD, and questions related to the *challenges in teaching* for the IS course. The characterization questions were generated from the attributes of the professors' profiles presented in the Brazilian formation guidelines for IS courses, given the continental size of Brazil and to cover its diversity [25]. The questions related to challenges in teaching were generated from a survey research performed in SE education from the professors' perspective with a similar purpose [12]. The authors invited six Ph.D. professors from different universities to answer the pilot questionnaire to check any issues before the study execution. After analyzing these answers and the professors' *feedback*, the authors made some adjustments to improve the understanding of the questions. The final version of the questionnaire consists of three sections: the first section presents the context of the survey and the consent form; the second section presents questions regarding characterization and challenges of teaching an IS course; and the third section consists of an acknowledgment for participating in this survey. The participant also had the possibility to inform his/her e-mail address to receive the results.

For the characterization questions, the authors defined closed questions (CQs) to identify the participants' profiles and support the quantitative results. Table I presents the list of CQs. For questions related to challenges in IS course teaching, the authors applied open questions (OQs). Then, the participants could describe their opinions more deeply. Additionally, such OQs allow these professors to expose their own difficulties and experiences in the teaching process in IS courses. The list of OQs is presented in Table II.

B. Execution

The survey was performed from September 18 to October 4, 2019. The authors chose such a strategy in order to guarantee

TABLE I
CQs

CQ	Closed questions
CQ1	What is your academic background?
CQ2	In which sector do you work?
CQ3	What IS course shifts (dayshift or nightshift) do you teach?
CQ4	What type of institution do you teach?
CQ5	What IS topics do you teach?
CQ6	Do you work or have you worked in the IS industry on the same IS topics you teach in the IS course?
CQ7	In general, do you teach at small classes, small-medium classes, medium classes, medium-large classes, or large classes?
CQ8	In which Brazilian state do you teach in the IS course?
CQ9	Is the city where you teach in the IS course small, medium-small, medium, large, or metropolis?

TABLE II
OQs

OQ	Open questions
OQ1	What subjects do/did you teach in the IS course?
OQ2	Which are the main difficulties you face when teaching to the IS course?
OQ3	If you teach subjects to other Computing and Informatics courses, do you face the same difficulties you listed in the previous question? Which are the other courses?
OQ4	Which IS topics do you consider most difficult to teach?
OQ5	Which characteristics do you think a professor must have to teach an IS course?

that the sampling was comprehensive. Due to not having an adequate and complete list of the subjects to be invited (IS professors), the authors applied multistage sampling [26]. First, the authors sent out the questionnaire to members of the Brazilian mailing lists from SBC: sbc-l² and sbc-si-l.³ They chose these lists because they are the largest and main means of dissemination among researchers and practitioners of computing and ISs in Brazil. As these lists have people who are not professors in IS courses, it is not possible to estimate the number of IS professors to obtain a precise response rate. However, as the participants cover all Brazilian regions and come from cities of different sizes, the authors consider that there was comprehensive geographic coverage. Second, they analyzed data from respondents who self-reported to be IS professors. This way, they tried to create an adequate sampling.

The questionnaire received a total of 70 responses, and all data collected were included in the analysis. The majority of the respondents hold a doctorate degree and currently work exclusively in academia, primarily in public universities. They teach a variety of subjects in IS courses, which are further detailed in Section IV-A.

C. Data Analysis

The authors applied descriptive statistics to quantitative results in order to summarize and rank the values from CQs.

²SBC mailing list: <https://grupos.ufrgs.br/mailman/listinfo/Sbc-l/>.

³Brazilian ISs Community mailing list: <https://grupos.ufrgs.br/mailman/listinfo/Sbc-si-l/>.

TABLE III
PARTICIPANTS' ACADEMIC EDUCATION

Academic education	Number of responses	Percentage
Doctorate \ Postdoctoral	53	76%
Master	16	23%
Graduated	1	1%

One researcher made the data entry. Next, the other researcher executed the descriptive statistics.

For qualitative analysis, the authors applied two steps of the coding process from the grounded theory (GT) Method. The GT coding process can be divided into three stages: 1) open; 2) axial; and 3) selective coding. According to Strauss and Corbin [23], although the GT Method purposes are the substantive theories' construction, the researcher may apply only some of its procedures to reach one's research goals. In this research, the authors applied open and axial coding. The open coding involves the data's breakdown, analysis, comparison, conceptualization, and categorization. After the open coding, the axial coding examines the categories, creating cause-effect relationships between them. The research question is broad and open-ended, and the literature may be consulted to enhance theoretical sensitivity [23], [27].

In this study, the second author generated a file containing the question and the corresponding answer for each participant. Then, the authors conducted two rounds of data analysis. The first and second authors performed the open and the axial coding individually for each round. Next, the axial step results were compared, and they tried to reach a consensus on the differences, including applying the kappa coefficient. They consulted the third author when there was no consensus in a code. The third author has experience in qualitative studies. After getting an agreement, the authors represent the results in graphical models (Figs. 8–10). The other authors validated these models.

IV. RESULTS

After carrying out the survey, the authors organized and analyzed the data quantitatively and qualitatively, as explained in Sections IV-A and IV-B. The authors discuss the threats to validity to these results in Section IV-C.

A. Quantitative Analysis

The quantitative analysis of the survey describe the participants' profile regarding the CQs presented in Table I and question OQ1 from Table II. The questionnaire received a total of 70 responses.

Table III shows the data related to the participants' academic education who answered the survey. Results analysis shows that most of the interviewees have a doctoral degree (76%), wherein part of them (23%) also carried out postdoctoral research. This percentage is the same of others who hold a master's degree.

Another factor evaluated in the survey was the sector in which the participants work. The authors observed that, among the respondents, the vast majority (83%, 58 respondents) work

TABLE IV
DISTRIBUTION OF PARTICIPANTS PROFESSIONAL PROFILE

Sector	Number of responses	Percentage
Academia	58	83%
Software industry	2	3%
Academia and software industry	9	13%
Governmental organization	1	1%

TABLE V
HIGHER EDUCATION INSTITUTION CATEGORIES

Institution	Number of responses	Percentage
Public University	44	63%
Federal Institute	8	12%
Public College	7	10%
Private University	6	9%
Private College	3	4%
Public and Private	1	1%
Research and Technology Foundation	1	1%

TABLE VI
PARTICIPANTS' PROFESSIONAL EXPERIENCE IN THEIR TEACHING AREA

Previous experience in industry	Number of responses	Percentage
Yes	50	81%
No	12	19%

only in academia; the second largest portion (13%, 9 respondents) work in both academia and software industry; and a few currently work only in the industry (3%, 2 respondents) and government organizations (1%, 1 respondent). This analysis is presented in Table IV.

In order to characterize the educational institution according to its academic organization and administrative category, the participants' answers were categorized as presented in Table V. The results show that most of the respondents (63%) work in public universities in the country.

The authors also evaluated the participants' performance in the software industry within the areas they teach in IS subjects. As evidenced in Table VI, it was observed that the vast majority (81%) works or has worked in the area of the taught subject, providing the professional in training with a better vision of the field, both in the theoretical and practical aspects. Seeking a better understanding of this factor, they also analyzed, among these professionals who work in the industry within their teaching area, which type of educational institution they belong to. The results show that most of these professionals work in public universities (54%), followed by Federal Institutes (14%).

Aiming to characterize the IS courses' profile, the authors evaluated the course class time (morning, afternoon, evening, and hybrid). The majority work in the evening time (39) or in the afternoon, and evening time (12), as shown in Fig. 1.

OQ1 question enabled the participants to list the subjects they taught in the IS course. The authors categorized the mentioned subjects according to the SBC reference curriculum⁴ [25] during the analysis. Thus, it was possible to identify the quantitative data regarding the participants' most taught group of subjects. The chart in Fig. 2 shows the number of mentions to each group of subjects. The group of subjects

⁴It is worth mentioning that the SBC reference curriculum is aligned with the ACM curriculum.

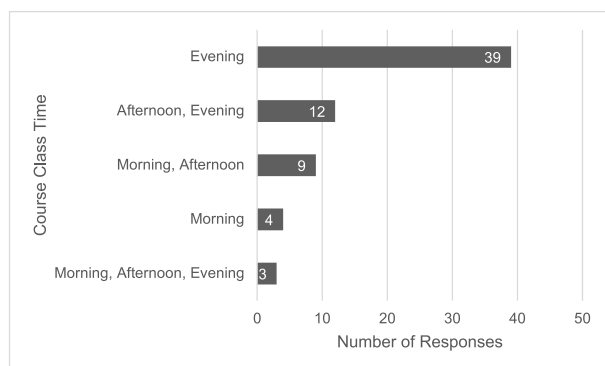


Fig. 1. Course class times informed by participants.

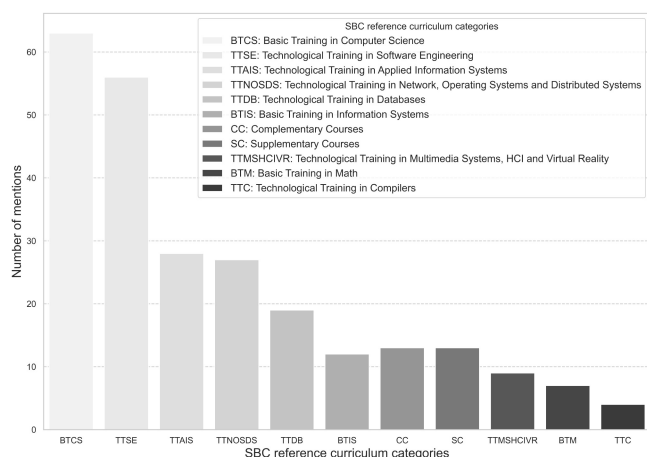


Fig. 2. Subjects from the answers grouped into the SBC reference curriculum.

that composes *Basic Training in Computer Science* was the most cited, with 63 mentions. Next, the group of subjects that contains the *Technological Training in SE* was mentioned 53 times. It is also worth highlighting the group of subjects that compose *Technological Training in Applied ISs* and *Technological Training in Computer Networks, Operating Systems, and Distributed Systems*, with 28 and 27 citations, respectively. The group of subjects that composes the *Humanistic Training* did not receive any mention and, thus, is does not appear in the chart.

The class size in which the participants teach was also analyzed. The results presented in the stacked bar chart shown in Fig. 3 indicate that class size mostly tend to be medium (20 to 40 students) with the percentage of mentions in the categories often and always reaching 63% of the participants answers. However, class of all sizes (categories) were reported by the participants.

Finally, the state and size of the city where the participants teach were addressed. The geographical distribution of participants who answered the survey can be seen in Fig. 4. Regarding city size, most participants teach in metropolitan-type cities (1+ million inhabitants), as shown in Fig. 5.

B. Qualitative Results

The authors performed a qualitative analysis based on the answers obtained in the OQs OQ2–OQ5 of Table II.

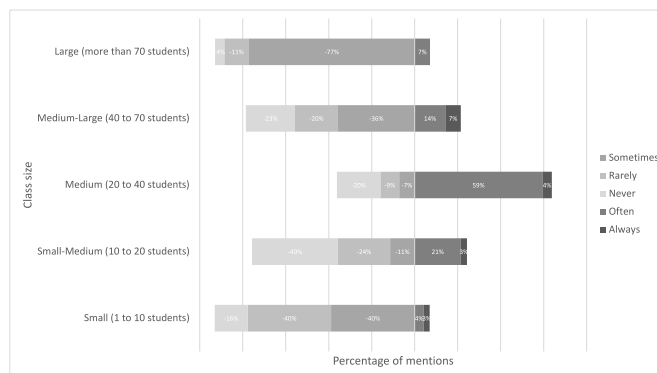


Fig. 3. Chart with participants' class sizes.



Fig. 4. Geographical distribution of participants.

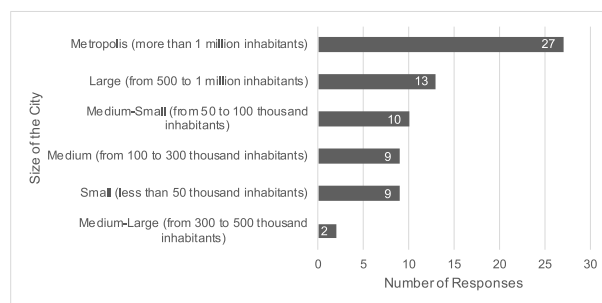


Fig. 5. Size of the city where participants teach.

1) *Main Difficulties in Teaching IS*: OQ2 sought to understand the main difficulties faced by the professor when teaching subjects to the IS course. In the qualitative analysis, the authors identified the categories: 1) difficulties related to the professor; 2) difficulties related to the students; 3) difficulties related to the course; and 4) difficulties related to the university. Each category also has subcategories, as shown in Fig. 6.

Furthermore, according to participant P21, “most faculties are trained in CS [Computer Science] and think IS is a weak CS.” Other participants also claimed difficulties in preparing classes that involve very specific content about which they have no experience.

Difficulties related to the teaching approach is another subcategory. Here, the participants reported that they have



Fig. 6. Mind map with the identified categories and subcategories related to the difficulties faced by the professors.

problems in adopting a teaching method that promotes greater student participation and adapts to the student's way of learning, as evidenced by part of the answer of participant P7: “[...] bringing teaching approaches that encourage a more participatory attitude from the students, with greater use of the student's presence in the classroom.” In addition to these two subcategories, other difficulties regarding the professor were the high number of students to supervise, teaching late at night, lack of ethics among fellow faculties and professors who do not profoundly participate in the university.

The difficulties related to the students are composed of the subcategories: 1) insufficient basic training; 2) lack of self-identification with the course; 3) lack of interest; 4) lack of understanding of the teaching–learning method; and 5) other difficulties. Regarding insufficient basic training, several participants claimed that students lack basic concepts from high school and essential IS course subjects. Participant P18 states that “Students arrive at higher education with weak basic education, having difficulty expressing ideas in the written form.” Participant P11 also reports that “basic concepts usually have to be remembered in class, taking part of the time dedicated to the content of the actual subject.” Professors also reported the students' lack of self-identification with the course and the subjects as one of the difficulties related to students. Participant P37 reports that “Students usually say that they are in the course just to have a higher education degree and that the score they got [during the selection process] ‘was enough to be admitted’ in the IS course.”

Another challenge reported by several participants is the lack of interest by the students. As described by participant P5, “Most students work during the day and go to the class tired and sometimes unmotivated [...]”, making it difficult for professors to keep students motivated. Participant P3 also reports “little student interest in theoretical subjects [...]”. In turn, participant P15 also claims that “students have little interest in programming,” which involves more practical aspects.

Regarding the lack of understanding of the teaching–learning method, participants reported that students sometimes seem unwilling to learn theoretical concepts before practice; some even want to perform their internship ahead of schedule. Furthermore, as exposed by participant P50, students have

“[...] the perspective the professor as the only holder of knowledge [...].” Participant P20 also reports the difficulty in “[...] Show to the students the difference between IS and other courses [...].” Other difficulties are related to students' distraction with technologies, such as the Internet, excessive use of cell phones, and the low attendance of students in classes.

Difficulties related to the IS course were classified into three subcategories: 1) difficulties related to the classes; 2) the course material; and 3) the IS teaching approach. Regarding the classes, the participants mentioned the difficulties of the course class happening in the evening, having huge classes, and having heterogeneous knowledge. The difficulties related to the course material were the lack of bibliographic references in Portuguese and the unavailability of material to carry out the activities at home. Among the difficulties related to the IS teaching approach are the absence of a broad discussion in IS teaching, using free practical examples in the classroom, and, according to participant P55, “[...] lack of a systemic approach to IS teaching.”

One of the subcategories that compose the difficulties related to the university is location. Some participants reported that the place where the course is offered is very far from the host institution; others said that there is a lack of companies in the city that absorb students for internships, as evidenced by the response of participant P51: “City, where the course is based, does not absorb students for internships, bringing the need for students to move to other cities in their course final periods.” Another subcategory is lack of resources. Participants responded that there is a lack of infrastructure for laboratories, teaching materials, investment for a professional qualification, services, and technicians that are not always available during class times. Participant P29 also reports the “Lack of specific laboratories for the SE area.” In the subcategory other, participant P12 describes that “Lack of support from the university for teaching activities (there is great pressure for research).”

2) *Similarity Among the Difficulties Faced in Other Computing Area Courses:* In question OQ3, the participants were asked about the recurrence of difficulties faced among the different computing courses in which they work. The aim of this analysis is to highlight the problems faced in the IS course regarding the other courses.

Considering the difficulties found in IS compared to other computing area courses, the authors can highlight the following categories: 1) courses that present the same difficulties faced in the IS course; 2) courses whose difficulties faced are similar but occur in less intensity; and 3) courses whose difficulties types differ from IS.

It is important to note that, during the analyses, some conflicts were identified among the answers. The conflict consists in the fact that some participants state that they face the same difficulties as IS in a given course, and other participants state that they do not face or face, to a lesser extent, the same difficulties for these same courses, such as Computer Science, Computer Engineering, and SE courses. The participants' answers were considered in the data analysis process in both cases, and the conflicts are indicated in Fig. 7 by the word “contradicts.”

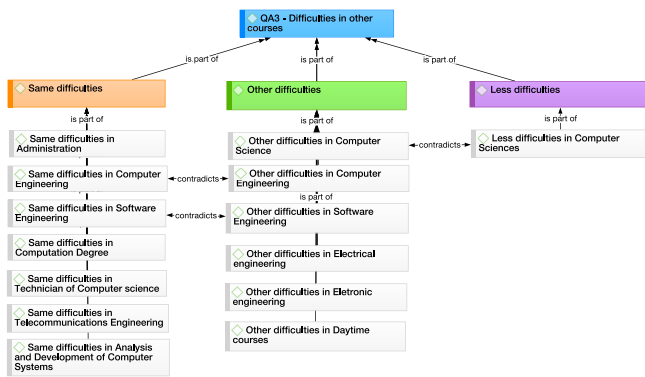


Fig. 7. Codes related to difficulties to teaching in IS course.

Within the courses category that show the same difficulties faced in the IS course are Business, Analysis and Development of Computer Systems, Computer Science Technician, Systems Analysis and Development, Telecom Engineering, Computer Science, and SE. The largest portion of the participants only claims to face the same difficulties. Some participants point out which difficulties are recurrent in some courses, as exemplified by participant P24 statement: “Yes, in the Analysis and Development of Computer Systems course there are difficulties which is also an evening class course. The students work during the day and study in the evening, and for this reason, they come to class tired and discouraged.” P65 and P57 corroborate with P24 and they add, respectively, “Yes, the material is available to do classroom activities. In fact, this difficulty is pertinent to any teaching area,” and “Yes, updated textbooks in Portuguese, lack full study time for the student. I would add, to the SE course, the lack of credit given by students from subjects that involve theory. They believe those subjects are dispensable.”

In the courses category with similar difficulties, but they occur in less intensity than IS, most participants stated that they face difficulties to a lesser degree in Computer Engineering and Computer Science courses. Some participants reported specific points where difficulties can be identified. For undergraduate courses, the main difficulties are related to students’ lack of motivation and preparation. In this context, the authors can highlight the following statements P3, P30, and P22, respectively: “computer science students sometimes show more interest,” “computer science students do not skip classes so much,” and “Only lack of motivation, but to a lesser degree. Course: Computer Engineering.” Participant P51 raises another important point relating to the subcategory of difficulties related to the university (OQ2), more specifically its location: “Yes, but with less impact, because there are some possibilities of internship in the region for bachelor degree in Computer Science.”

Finally, the last category encompasses the courses with different difficulties from IS. Among these courses are Computer Science, Computer Engineering, Electronic Engineering, Electrical Engineering, and SE. Although most participants only state that they do not face the same difficulties, some participants report what these difficulties are. The main one is that students from dayshift class courses do not face the same

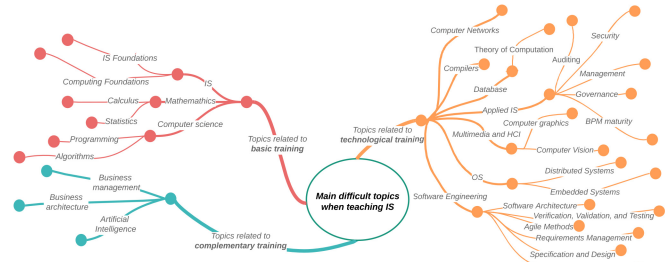


Fig. 8. Topics that are difficult to teach in IS courses.

problems presented in IS courses. Participant P5 states that “Students arrives better prepared in dayshift class courses.” Participant P38 emphasizes that “[...] because they are dayshift class courses with most students having exclusive time for graduation.”

3) *IS-Related Topics That Are Difficult to Teach:* In order to identify the IS-related topics that are difficult to be taught during the course, in question OQ4, the participants were asked about their point of view on this aspect. The results were characterized and divided among the main areas of IS knowledge, as presented in Fig. 8. The areas that involve the subjects with the greatest difficulty in IS teaching are: 1) areas related to technological training; 2) areas related to basic training; and finally 3) areas related to complementary training. Within each area are the main topics and subjects classified as difficult to teach by the survey participants.

In the category areas related to technological training, the authors grouped themes related to technological training in applied training. This subcategory involves the topics of systems auditing, systems management, systems governance, BPM maturity, governance, and information security. Participant P52 exemplifies this subcategory: “Themes related to business and systems organization concepts.” In this category are still the subcategories: compilers, with the themes compilers and theory of computation; database; and multimedia systems and HCI, with the themes computer graphics, Human–Computer Interface, and Computer Vision. Another subcategory involves Computer Networks, OS, distributed systems, and embedded systems as more difficult subjects to teach, as stated by participant P13: “Implement and operate infrastructure involving computer networks, cloud, cluster, etc.” The subcategory SE was mentioned many times among the participants, involving the topics: software architecture, requirements management, agile methods, systematic methods of specification and design of ISs, software metrics, design and analysis of systems, requirements verification, and software testing.

In the category of areas related to basic training are the subcategories of basic training in computer science, mathematics, and IS. Within this category, the authors can highlight IS fundamentals, computing fundamentals, graphs, topics that involve logical and computational thinking, calculus, and statistics, prevailing the topics that involve basic training in mathematics, as stated by the participants P5: “[...] the part of data analysis demands a greater mathematical students’ knowledge. I find this part more difficult to make them

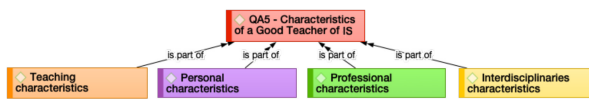


Fig. 9. Codes related to characteristics of a good professor in IS courses.

understand,” and P11: “*topics like mathematical proofs and formal equations representation are confusing to students with weak mathematical background.*” In the subcategory of basic training in IS: the IS fundamentals subject and Computing foundations were cited. The subcategory of basic knowledge in computer science comprises the subjects of algorithms and programming primarily. However, related subjects, such as data structure, algorithm analysis, and algorithm complexity were also cited.

In the category of related areas to complementary training, involving complementary subjects to the course, the subjects involving business/management, business architecture, and Artificial Intelligence were cited.

4) *Characteristics Required From a Good Professor in IS Courses:* In question OQ5, the participants were asked about the main characteristics inherent to a professor that are essential for the good conduction of subjects in IS courses. According to the results obtained and shown in Fig. 9, four main categories were identified: 1) personal characteristics, related to the professor’s personality and attitudes; 2) professional characteristics, related to academic and professional experiences; 3) teaching characteristics, related to the professor’s didactic; and 4) interdisciplinary characteristics, related to the professor’s domain on other knowledge areas.

Considering the category of the professor’s *personal characteristics*, these are the characteristics expected of a good professor: to be patient, to enjoy teaching, to possess ethical and professional attitudes, innovative spirit, interest in teaching, leadership, to know how to demand from students, to be attentive, committed, communicative, creative, dedicated, dynamic, empathetic, flexible, motivated, proactive, objective, resilient, to create his/her own material and keep it updated.

In *professional characteristics* category, the following characteristics were cited by the participants: keeping aligned and updated with the IS market and new technologies; having scientific, technical, theoretical, and practical knowledge; knowledge in areas related to IS, psychology, and psychoanalysis; having technical, technological, and pedagogical training, especially, in the subjects taught; having previous experience in the labor market, mainly in system development and management. In this category, the authors can highlight the professor’s experience in the labor market as a desirable professional characteristic, as stated by the participants P65: “[...] preferably have the market experience to be able to show in practice some more theoretical subjects,” and by P66: “*Preferably have the market experience to bring to the classroom.*”

In the *teaching characteristics* category, the most cited characteristic was *present good didactic*, cited by seven participants, followed by the ability to exemplify the content with real-world problems, as stated by participant P2: “*It is important to bring to the classroom practical situations from the*

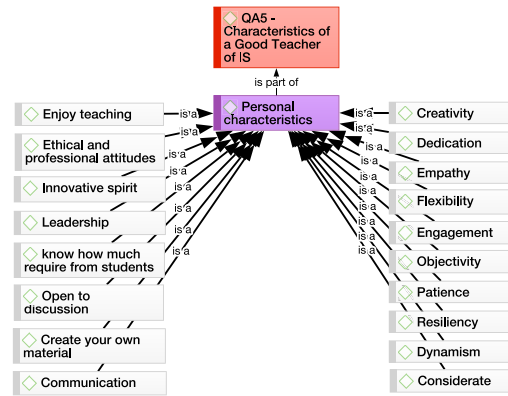


Fig. 10. Expansion of personal characteristics of a good professor in IS courses.

real world,” and by P40: “[...] look for real problems for students to get more involved.” In addition, other characteristics were also cited: the ability to adopt innovative methodologies, add a monitors team, apply content to practical problems, engage students, manage group work, promote active teaching, reflection, and critical sense of the student, have available time, diversify content, avoid content-based and script-driven teaching, diversify teaching methods, involve students in class dynamics, be able to adapt to how students learn, and diversify practical activities.

In the *interdisciplinary characteristics* category, the participants report the importance of the professor being able to establish an interdisciplinary vision among the areas that involve their subject, as stated by participant P2: “*The professor tends to know other areas and, thus, facilitate the interdisciplinarity which is so important in IS.*” Some participants relate interdisciplinarity with market vision, as cited by participant P55: “*A ‘market vision’ is also essential for the formation of competent professionals aligned with professional demands.*” In addition, the participants report on the importance of: 1) having practical, critical, humanistic, and interdisciplinary vision; 2) having management vision of the labor market, curricular matrix, and general solutions; 3) being able to identify repeatability in the curriculum; 4) having the ability to understand interdisciplinarity, and local industry needs; and 5) bring new technologies and understand that IS is a technological artifact inserted in an organizational context. In addition, the professor must be aware of the transformations in society brought by Information and Communications Technologies (ICTs).

In order to exemplify how the characteristics are arranged within the main category, Fig. 10 illustrates the visualization in the Atlas TI tool of the category expansion, personal characteristics, and the individual codes representing each characteristic cited by the participants, as listed above. Other categories and their components can be explored in the repository available online at: (https://github.com/vaniaon/si_education).

C. Limitations and Threats to Validity

Similarly to other survey researches, there are potential threats to the validity of this study that the authors should

consider herein. First, there is a risk that the participants do not understand the questions. As such, they conducted a pilot with a limited number of participants who gave to them feedback on the questions to mitigate this risk.

As the questionnaire was publicly available, it is possible that people who do not work as professors or people who are professors but do not teach for IS courses have answered the questionnaires. To prevent this, the authors emphasize in the questionnaire description that the target audience is faculties that teach IS courses.

It is possible the survey respondents do not represent the entire population of professors that teach IS courses in Brazil. To reach a broader and more diverse population, as explained in Section III-B, the authors disseminate the questionnaire on the sbc-l and sbc-i-l mailing lists that are well known in the Brazilian scenario via SBC.

Since qualitative analysis is a subjective method, which can lead to different interpretations, to mitigate it the first and second authors perform it parallelly and independently. Then, they compared both analyses and discussed to reach a consensus. When there was conflict, the third author was in charge of solving it.

The research was conducted before the Covid-19 pandemic. Then, the results reflect the professors' perception in that context.

V. DISCUSSION AND IMPLICATIONS

Based on the results presented in Section IV, this study contributes to the existing knowledge by reporting several difficulties IS professors face in their work. The main classes of difficulties/challenges raised in the results are *Challenge 1) IS Course* (teaching approach, material, and classes), *Challenge 2) University/Institution* (mainly location and resources), *Challenge 3) Professor* (experience and knowledge, alignment with theory and practice, and teaching approach), and *Challenge 4) Students* (lack of understanding the teaching method, lack of interest and self-identification with the course, and insufficient basic knowledge). This section revisits these challenges and explores possible ways to overcome them. The discussions also consider the implications and impacts of these actions on the practice of IS education. A comparison with existing studies is also performed to evolve the state-of-the-art.

A. Adopting Teaching Methodologies That Motivate Students

This action can contribute to *Challenge 1 (IS Course)* and *Challenge 4 (Students)*, referring to the adoption of teaching approaches to foster students' participation. This approach has been widely acknowledged in the academic community and is corroborated by [9], [10], and [19]. Studies advocate for the use of student-centered approaches, e.g., the flipped classroom [10], problem-based learning [11], and industry-based projects [20], which have the potential to enhance student motivation and learning outcomes—an important implication. In fact, a pedagogical approach using problem-based learning and dynamic classes resulted in increased student motivation and engagement compared to traditional, theoretical

assessments [11]. However, implementing student-centered approaches can pose challenges [19]. The additional workload required for these methods can be a burden for students who already face the demands of working and studying, which may hinder their engagement in activities that require active participation.

B. Getting Closer to the Industry

The lack of approximation with the industry was mentioned as a difficulty faced by professors and also related to the course. This action can majorly contribute to *Challenges 1 (IS Course)* and *3 (Professor)*. Gill and Hu [15] supported the notion that professors should have both a comprehensive understanding and hands-on experience in the field they teach to ensure alignment with industrial practices. According to the authors, the technical competence of professors in their teaching area and nonacademic professional experience related to IS are crucial aspects that define faculty quality. The approximation with the industry also favors bringing real problems to the class and adopting innovative technologies, which was a need highlighted in the present study. Sahin and Celikkan [8] corroborated this result and emphasize that, in order to minimize the gap between the existing Information Technology education system and the industry's needs, rather than theoretical emphasis, courses should include practical projects that meet real needs of the sector.

In this study, it was not possible to determine a conclusive correlation between the number of people working in the industry besides being a professor and the Brazilian states from which they come. For instance, Rio de Janeiro, Amazonas, and São Paulo had the highest number of professors who also work in the industry, in that order. Nevertheless, it is important to note that this data could be biased by the number of respondents from these states, which were among the most highly represented in the study.

The establishment of closer ties between academia and industry in Brazil can be achieved through two strategies. First, by promoting the recruitment of professors who possess prior industry experience and/or by employing part-time professors who are currently active in the industry. While full-time professors are typically hired for 40-h work per week at Brazilian public institutions, reserving some positions for 20-h shifts could enhance the faculty, and foster the exchange of experiences between full-time professors. This is particularly important as many IS courses in Brazil are taught during evening hours. Second, by encouraging the development of university–industry partnerships to leverage real-life cases in the classroom, foster innovation, and help produce well-rounded professionals. The implications of such actions are significant. A faculty that is well equipped with relevant industry knowledge and experience can provide students with a more practical and up-to-date education. This, in turn, can increase the value of their degrees and better prepare students for their careers in the field of IS. Partnerships can benefit both academia and industry by promoting a mutual and beneficial exchange of ideas and resources, leading to advances in both sectors.

C. Improving Professor's Training

This action can majorly contribute to *Challenge 3 (Professor)*, as developing new teaching methodologies that motivate students is necessary. However, to achieve this goal, a professor must have a pedagogical background. The importance of pedagogical training was highlighted by the participants as characteristic of a good professor, which is consistent with findings in [13] and [14]. Therefore, it is crucial to promote actions that encourage pedagogical training, such as: 1) continuous training offered by the universities through short courses or even promoting events to exchange teaching experiences and 2) improving the training of future professors by introducing pedagogical foundations in the curricular components of undergraduate and graduate levels (optional at the undergraduate level).

The IS department could offer training and events to help professors and students keep up to date with the dynamic IT industry, as respondents reported difficulties in staying current with technologies and tools used in the field. Short-term workshops led by industry practitioners could also contribute to bridging the gap between academia and industry. This challenge was also identified by Ferreira et al. [12]. Interestingly, the work of Gill and Hu [15], conducted in the 1990s, did not report this problem at that time. Obviously, the current scenario is vastly different, with a greater diversity of technologies and a faster market pace being the reality today.

Participants report the importance of a professor being able to establish an interdisciplinary perspective on a practical, critical, and humanistic vision—in this scenario, respondents have cited knowledge in areas related to IS psychology and psychoanalysis. The latter is supported by the study of Matturro and Solari [7], which identified professional characteristics in areas related to Psychology and Psychoanalysis among those required from a good IS professor. In addition, social psychology is considered a soft skill within the interpersonal category. Sahin and Celikkan [8] also supported the notion that soft skills should not be ignored, and analytical thinking and teamwork are the two most requested by the software market.

The IS area is a convergence of Informatics, Business, and Social Sciences. An important result obtained from the 70 respondents is that *none of them mentioned subjects in the humanistic area*. The most likely reason is the lack of training to offer such a subject. A possible absence of these subjects is perceived, despite their importance in organizations, such as organizational communication. Moreover, such topics are usually “outsourced” in the Brazilian context, i.e., professors that often teach them are from other academic faculties at the same university. For instance, management schools are often responsible for offering subjects related to enterprise management for IS bachelor's degrees in Brazil.

Finally, the implications are two fold: 1) if professors that offer humanistic courses are from the IS faculty, they need proper training on these subjects and 2) if professors are from a different faculty, they may receive IS training to adapt the content of their course to reach IS students' needs. The training could be, for example, through instructional video courses to

help them better understand the area and develop/adapt specific strategies for IS course students.

D. Promoting the IS Course

This action can majorly contribute to *Challenges 1 (IS Course)*, *3 (Professor)*, and *4 (Student)*. The difficulties associated with teaching IS corroborate with reports from existing studies in [1] and [12]. Regarding the professors' experience and knowledge (*Challenge 3, Professor*), Lisboa [16], in a seminal work, had already stated that an important aspect is technical competence in the professors' areas and that nonacademic IS-related employment experience is an indicator of a university's quality. Other studies over the past two decades have also discussed how close IS professionals' skills are in relation to those found in newly graduated professionals [18], [19].

In addition, several studies investigate the reason why students choose an IS course, but none of them have explored the ease of entering such a course. This fact may be a characteristic of the admission process at Brazilian universities and requires further investigation [12]. However, since the current study is qualitative in nature, such cases may be isolated and not reflect the overall reality. Nonetheless, undergraduate IS managers should adopt strategies that promote and provide information about the IS course to high school students, as well as how professional activity works in this field. Zhang and Zheng [5] and Chipidza et al. [6] have suggested the development of informative content (*Challenge 1, IS Course, branch “content”*), e.g., websites and videos featuring former students, and organizing events and panels that promote interaction between the target audience and IS professionals. These measures can help high school students understand the course before entering university (*Challenge 4, Student, branch “student engagement and self-identification with the course”*) and engage with the IS course more effectively.

E. Strengthening the Basic Education Level

This action can majorly contribute to *Challenge 4 (Student)*. According to most respondents, a lack of basic knowledge, mainly related to mathematical skills, difficulty in interpreting problems, and text comprehension, are significant challenges students face. This difficulty is justified by students arriving at university without the necessary background required for the course [12]. This fact is supported by several authors who have identified similar problems in specific disciplines, such as programming [14], [28], [29], [30], or even in Engineering [31], leading to students dropouts.

Vieira et al. [30] found that students who attended a Technical Course in IT were more likely to engage in an IS course than others. Therefore, it is essential to reinforce students' primary and secondary education and include basic computer science education in reference curricula through national public policies. Introducing basic programming topics at these levels can promote the development of skills for IS and other computing courses, and students can be encouraged to consider computing careers, as supported by [32].

Scaico et al. [32] implemented an outreach project with high school students from a Brazilian public school, offering long and short courses as well as competitions related to computing. The authors found that the number of students from this school entering universities and selecting courses in the computing area had increased compared to previous years.

F. Creating Hybrid Courses and Remote Support

This action can majorly contribute to *Challenge 2 (University)*. As supported by other studies, students often face difficulties traveling to the campus where classes take place, which is usually far from urban centers [1]. Converting IS courses into hybrid courses emerges as a solution adopted in the pandemic, which involves teaching part of the courses remotely and part with in-person classes. This approach can reduce the effort and cost of commuting and staying at the university for at least some days of the week, including expenses, such as fooding and clothing. However, this shift also impacts the university routine and the students' experience, requiring planning, and evaluation [31].

The results of this study indicate that respondents found topics from three fundamental areas of IS knowledge (technological training, basic training, and complementary training) difficult to teach and requiring extra effort, such as remote supporting classes. To investigate this issue further, the authors analyzed the relationship between answers to questions OQ1 and OQ4. The analysis suggests that the subject being taught may influence the topic that a professor considers challenging. This factor could introduce bias, and additional data is needed to confirm its influence. However, a combination of remote supporting sessions and complementary monitors alongside traditional classes can provide additional support to professors and improve student engagement, as observed by Ferreira et al. [12].

VI. FINAL REMARKS

The main contribution of this study was to report on the challenges faced by IS courses in Brazil as perceived by professors. The IS degree is the only one computing-related career that accordingly prepare the professional for effectively communicating with both the nontechnical CEO of a company as well as with the programmers, being the glue that connects technology, social, and business dimensions into the corporate environment. It is then essential to recognize the challenges faced still in the undergraduate and reinvent the career. After the Covid-19 pandemic, higher education is pressured to deliver professionals that can bridge the technical and non-technical worlds and arise innovative solutions for the market. The results communicated in this article can serve as input to guide the establishment of new strategies and directions for IS degree formation in Brazil, enabling it to be reinvented to face the current and forthcoming challenges.

The results lead to important implications for the teaching process in IS courses in Brazil.

- 1) Actions for continuous training offered to IS professors by industry practitioners sharing knowledge on tools currently used in industry.

- 2) Establishment of institutional policies to support students in several difficulties faced in their courses based on an academic sector or department to provide extra-class support in some critical topics/subjects, background classes for new students, opportunities for getting different types of scholarships (research, internship, social actions etc.), and psychological and pedagogical support from professionals who are aware of the IS courses particularities.
- 3) Creation of hybrid courses with in-person and remote classes to reduce the barriers to students, especially, considering the fact most of professors report that IS students are working either in the field or for support their families, besides issues related to traffic jam, distance from home/work, and transportation.
- 4) Course class time offering in the mornings and afternoons for those who can (or intend to) be dedicated to an IS course, similar to other Computing and Informatics courses in Brazil (e.g., Computer Science, SE, and Computer Engineering).

From these implications, this study opens opportunities for further research. First, it is important to develop a framework for continuous training of IS professors, who are not necessarily grounded in an IS course, considering the interdisciplinary nature of an IS course, which comprises subjects on Informatics, Business, Social Science, and Math. In this case, some characteristics should be worked as identified in this study, such as innovative spirit, creativity, dynamism, communication, and engagement. Second, a method for continuous updating of the IS reference curriculum should be developed since the area comprises competences joint from the people, processes/organizations, and technologies dimensions. This is important to avoid misunderstanding the particularities of an IS course against other Computing and Informatics courses. Finally, strategies to integrate different subjects in the same semester based on real projects from partnerships with some university sectors, companies or government should be proposed and evaluated. This is critical since both IS professors and students are learning and updating themselves daily.

In addition, a similar study can be conducted to allow a diagnosis of difficulties faced by students in IS courses in Brazil and compare the results with the professors' opinions. Once the research considered the context before the Covid-19 pandemic outbreak, further opportunities include replicating the research with professors *post*-pandemic phenomenon and also replicating it with students. Moreover, the establishment of a roadmap for coping with the difficulties identified in both studies, considering the strategic importance of IS courses as instruments to sustain Computing and Informatics in the industry in different sectors.

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